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SECURE INFORMATION EXCHANGE IN ORGANISATIONS *H7***

Secure Information Exchange in Organisations

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Abstract:

This paper describes security criteria as components of a comprehensive security framework for information exchange in organisations. The proposed security criteria correspond to the basic threats to information exchange and the fact that information is a valuable resource in organisations. According to these threats more complex security requirements arise which demand traditional IT-security criteria to be extended. The principles of the extension and the interaction with traditional approaches are demonstrated.

Keywords:

Authentication, Availability, Confidentiality, Goal Conformity, Information Exchange, Information Security, Integrity, IT-Security, Obligation, Security, Security Criteria.

1. Introduction

Information is regarded as a vital resource for every kind of service and function provided by organisations. The division of labour in organisations requires that information must be exchanged between communication partners in and between the organisation. Security then becomes a major question as the various threats concerning the dependency of the organisations to the resource information must be dealt with. In this paper we propose a security framework for information exchange. In doing so, we explain the term security and the characteristic properties of information exchange in organisations. By analysing different dimensions of information exchange in organisations we show the distinction between what is traditionally called data security and what we call information security. Finally, we define security criteria relevant for IT-systems which are used for the exchange of information in and

between organisations. We describe these criteria as a components of a comprehensive security framework¹.

The security framework is a contribution to the construction of secure IT-systems because we extend traditional IT-security criteria (loss of confidentiality, integrity, availability) which is a precondition for the development of comprehensive security concepts. This extension results from a systematic analysis of security requirements for information exchange in organisations. The paper is structured as follows: In chapter 2 we first explain the term security and derive a definition for this term. After that we introduce three basic dimensions of information exchange (chapter 3). These dimensions emphasise the economic relevance of information and the interdisciplinary nature of information exchange in organisations. They show the inseparable relationships between the exchange of information and its subsequent use. As an implication the traditional IT-security criteria need to be re-examined (chapter 4). Information security requires the recognition of how information is used in planning and decision making processes and how it effects the activities of the communication partners. We call this interaction the economic dimension of information exchange.

In chapter 5 the components of our security framework will be described. The foundation of this framework is formed by three basic dimensions of information exchange. Due to this structure we derive the corresponding security criteria as the components of the framework. In the subsequent sections we discuss each of the criteria within the framework followed by some important remarks (chapter 6).

Finally (chapter 7), we give an overview of the functionality which an information system needs to realise in order to secure information exchange.

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2. Security is an abstract concept

Security is an abstract concept and as such has no physical representation in the real world (there is no physical object called security). This makes it more difficult to achieve a common understanding of what security means. Since there are no external representations of security, only individual imagination can be considered (i.e. what people have in mind). Thus, a definition of security must deal with subjectivity [35].

Security is the property of a system² that can be defined as: threats considered as relevant and directed to specific valued objects, be addressed by specific measures so that the resulting risks are deemed acceptable [26]³.

In this definition subjectivity addresses the relevant threats, objects to be protected and the acceptance of resulting risks. The explanation of relevant threats must be based on assumptions of possible damage, i.e. the description of situations in which a secure system state breaks down [35]. It is impossible to give an exact or complete description of future situations, because their evaluation depends on many components which cannot be fully anticipated.

One prominent approach to handle the subjectivity of security is the definition of basic threats [29]. In traditional approaches to IT-security the basic threats are loss of confidentiality, integrity, and availability [8] [1]. With respect to communication system requirements these basic threats are extended by the loss of authenticity. This approach allows a structured analysis of individual security requirements based on the influence of the basic threats to valuable objects, i.e. risk analysis [4]. Further on, it is possible to develop generic services for securing such valuable objects.

However, the basic threats above cover the electronic part of information exchange and are independent of the purpose IT-systems are used for, i.e. they do not consider the application domain. Our approach to security for information exchange takes a more general point of view encompassing the IT environment, i.e. interpersonal relationships and the use of information by humans to support decisions and actions in the real world. The following chapters emphasise

² Not only of an IT-system.

³ This definition is a result of the REMO project.

the problem of security for those parts of information exchange which are inseparable from IT-use but not covered by the IT-system. We will review the basic threats of IT-security by analysing the characteristics of information exchange in organisations. As a result this analysis shows the need for an extension of traditional basic threats to meet the requirements of this application oriented perspective to security.

3. Information and Information Exchange

In order to identify basic threats and valuable objects for information exchange we will first introduce our understanding of information and explain the necessity of exchange to generate information. It enables us to differentiate between relevant terms for doing this analyses which are frequently confused: *information*, *knowledge*, *data* and also between *information exchange*, *content exchange*, and *data exchange* (Fig. 1). These terms reflect the basic dimensions of information exchange.

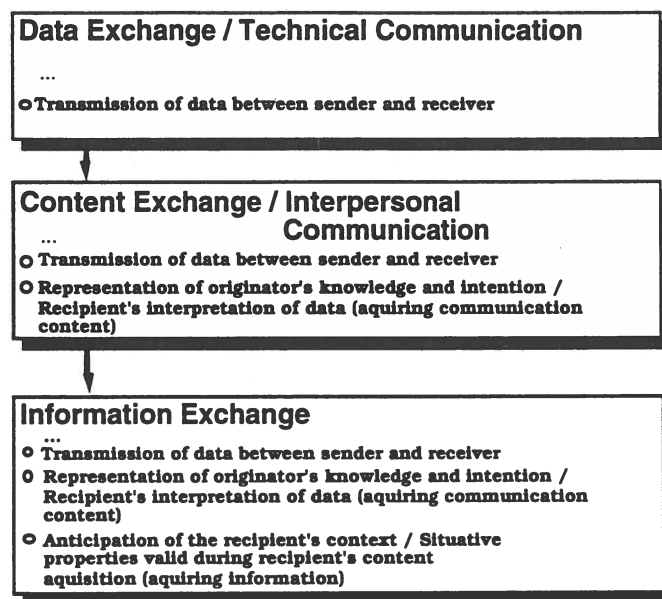


Fig. 1: Data exchange, content exchange, information exchange

3.1 Dimensions of Information Exchange

The fact that there is a division of labour in and between organisations necessitates the exchange of information between communication partners. A generic structure of information exchange in organisations consists of three dimensions (Fig. 2): social interaction, economic potential, and technical data exchange. The technical dimension and the social dimension are established in interpersonal communication [35] [9]. According to the economic relevance of information

in organisations we introduce the economic dimension as the third dimension of information exchange in organisations. This dimension will be focused in the following chapter.

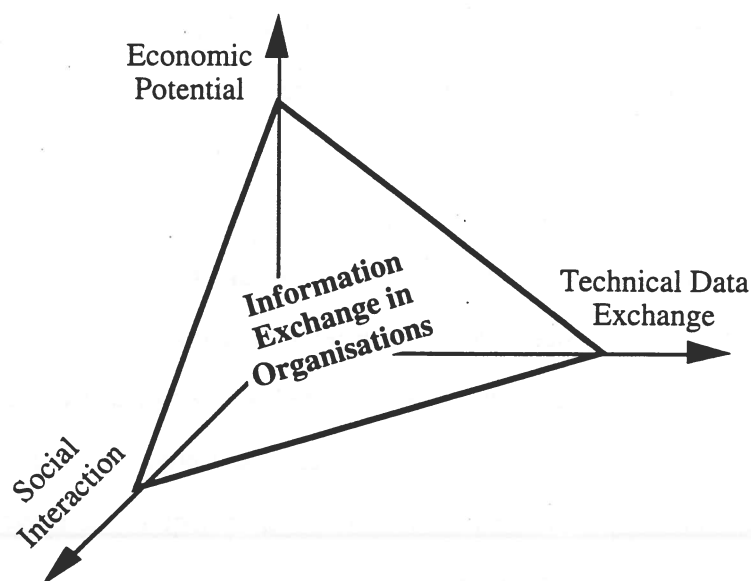


Fig 2: Dimensions of information exchange

Social dimension:

The social dimension consists of the interpersonal relationships within social interaction. Obligations and assertions between the communication partners are generated corresponding to the semantics of the interpersonal communication (content exchange in Fig. 1).

Technical dimension:

The technical dimension covers the transmission of data between the communication partners. Therefore, it is an implication of the demand for information exchange in organisations.

Economic dimension:

The economic dimension is built on the need for information in order to fulfil tasks in organisations (information is a precondition for task fulfilment). The economic potential given by information depends on the relevance for actual task fulfilment, i.e. the actual context of the communication partners (information exchange in Fig. 1).

3.2 The Economic Potential of Information Exchange

Usually, in the area of information system research, *information* is described as "goal-oriented knowledge". Goal-oriented knowledge implies that information must be useful for specific

purposes (goals). Buckland shows difficulties in defining the term information. He discusses the need for three different viewpoints: information as "process", "knowledge" or "thing" [5].

In the area of Information Science information is called "knowledge in action" (Kuhlen) [16]. This implies an integration of process view and knowledge view. However, information as "thing" is called data. An analogous differentiation between databases and corresponding information spaces is made by Schmidt and Bannon [4, 25].

Kuhlen's explanations of knowledge in action reflect the natural use of the term information including what "goal-orientation" means: When interpersonal communication takes place, *knowledge, facts or believes* of a sender must be transported to a recipient as *data* (encoded signals). The term *information* is used if several context dependent properties are valid within the process of reception and interpretation of the data in order to acquire the content, i.e. the semantics (knowledge, facts or believes). Information must

- be *addressed* to a recipient;
- be *interpreted* by the recipient (he or she must be interested);
- be *context relevant*: goal-oriented, timely;
- be *new*, i.e. not part of the recipient's current knowledge;
- have *pragmatic relevance* for the recipient (relevance for recipient's current activities).

The pragmatic relevance is one of the most important properties of information. This is confirmed by the following statement [19]: "Information is only useful because someone can do something with it, ...", (Medina-Mora, Winograd, R.Flores, F.Flores, p 282).

As shown in Figure 1 data exchange covers the technical part of information exchange: the transmission of data between communication partners. *Content exchange* additionally includes the semantics of the interpersonal communication, i.e. a transformation of the senders knowledge and intention to data representations and the subsequent interpretation process of the recipient to achieve meaning (content acquisition). *Information exchange* finally requires an intention of the sender (anticipating the recipient's current context) and situative properties to be valid within the interpretation process. These properties depend on the actual context and determine the informational value as relevance for the recipient's current activities. It is called the economic relevance of information and backed up by the dependence of the industrial world

on information which must be timely at the right place with a minimum of costs. Therefore, information can be viewed as a production factor similar to management, planning, organisation, control [34] [23]. It is a major factor in supporting the management processes and forms the basis for decision making . However, information cannot be delivered as a whole instead it must be acquired by a recipient out of some raw material (data). Information therefore becomes a special kind of economic good [18].

The significant property of information is that it enables a recipient to act in an organisational context (to fulfil a task, e.g. to make a decision). Goal-oriented and validated activities⁴ depend on the knowledge of the decision maker as well as relevant information which must be acquired as part of an information exchange. This is the reason why we call information a production factor, i.e. an economic, valuable good. Based on this understanding of the term information we introduce some new security issues in the following chapter.

4. Data Security versus Information Security

4.1. The Difference

The economic relevance of information in organisations and the distinction between data and information as shown in chapter 3 have implications for the definition of security requirements and as a consequence for the definition of security criteria [11]. It becomes clear that security requirements must not be restricted to the states and processes of the IT-system and the corresponding data. Since IT-systems are not only used to process data more efficiently and increase capacity, but are inseparable from the exchange of information between communication partners that supports people in goal-oriented activities (e.g. carrying out their tasks), new security requirements arise. Primarily the confidentiality of information must be redefined.

Traditional approaches to IT-security concerning confidentiality reduce this problem to the question of data security, i.e. prevention of unauthorised access or disclosure. Security measures consist of mechanisms that prevent against unauthorised data access and misuse of

⁴ A broad pragmatic understanding of the term activities includes physical and intellectual acts.

services for data processing, i.e. they enforce a well defined sequence of IT-internal system states [10]. Confidentiality is equated with data access controls and data encryption. The possible use and misuse of information which can be acquired is not considered.

We emphasise the difference between data and information as shown in the previous chapter. Therefore, *data security* concerns data access while *information security* concerns the use of information and depends on context dependent interpretation processes (acquisition of information).

If an organisation "owns" valuable information then an important security requirement is that this information is used to support the goals of the organisation when it is exchanged between employees and/or external partners.

4.2. Extension of the Basic Threats

Due to the distinction between data security and information security the corresponding basic threats must be considered. Therefore, acting humans must be involved because information exchange causes interpersonal relationships and influences the behaviour of these humans. Threats in this case are based on conflicts in goals and interests between human communication partners.

Traditional security concepts support the requirements for data confidentiality due to the use of IT-systems for data transmission, value added services (pre-processing of information⁵), administration and also automatic processing of data. The specification of rights and responsibilities for realising access controls are based on the questions: "What does an actor actually do? What must he do? What is he allowed to do?" Complex mechanisms are developed to realise such data access controls [15] [22] [17] [30] [33] [32] [13]. However, these approaches cannot influence the subsequent processes outside the IT-system: the intended use and possibilities of misuse of information.

⁵ In this state we can say potential information because a human interpretation process, necessary to validate the situative properties described in chapter 3, is not included.

This depends on alternate possibilities for acquisition and use of information derived from data accesses. The important questions are: "What possible uses of the accessed data exist? Which overall goals does it support? What kind of misuse is possible?" These questions show additional threats in case of authorised data accesses which are currently not covered by security mechanisms of information systems.

The following basic threats represent the additional security requirements for information exchange according to our understanding of information security. These basic threats extend traditional ones by including inadequate human behaviour caused by conflicting goals or interests, as described above:

- Misuse of information: The sender has to deal with threats resulting from an inadequate behaviour of the recipient (the acting person). Inadequate behaviour means that the information the sender gives to the recipient enables or causes activities which do not correspond to his goals or interests.
- Use of incorrect information: The recipient has to deal with the threat that he will be guided into the wrong direction. He wants to make valid decisions and engage in desired activities by using trustworthy (correct) information.
- Neglecting obligations: Each communication partner has to deal with threats resulting from inadequate behaviour of the other one. Inadequate behaviour means that communication leads to a social interaction [35] and causes obligations between the communication partners which they are unwilling to fulfil.

5. Comprehensive Security for Information Exchange in Organisations

5.1. Security Concerning the Dimensions of Information Exchange

In the previous chapters we explained the relationship between the exchange and subsequent use of information as well as the corresponding threats. We will now use these insights to derive a comprehensive specification of security criteria.

Security as we have defined in chapter 2 is a property of a system that refers to individual threats and valuable goods. Identifying basic threats provides an abstraction from this individual point of view. Therefore, we will now describe basic threats and valuable goods according to the three dimensions of information exchange and define the corresponding security criteria in the subsequent section:

Technical dimension:

The valuable good is the data transported between sender and receiver. The basic threats are unauthorised access or manipulation, i.e. a loss of "*Confidentiality, Integrity and Availability of the data and services*" or the "*Authenticity of the communication partners*".

Social dimension:

We define the obligations and assertions which are generated between the communication partners as valuable goods of the social dimension.

Neglecting obligations is the basic threat to these valuable goods. The binding character of obligations is lost, i.e. a communication partner can neglect obligations, if the "*Authenticity of the communication partners*" or the "*Agreement of the Content of the interaction*" is lost.

Economic dimension:

As a valuable good we define the pragmatic relevance of information. There are two important points of view concerning this valuable good:

- The use of information should be restricted to the intended goals of the sender.
- The information should build a valid basis for the permitted activities of the receiver.

Therefore, the basic threats are misuse of information or manipulation of behaviour, i.e. the "*Use of information bound to intended goals*" or the "*Authenticity of the content*" is lost.

Traditional IT-security primarily concerns the technical dimension. Therefore, confidentiality is reduced to data security, i.e. to prevent against unauthorised data access (disclosure) and the misuse of data processing services. In this sense it allows to enforce *confidentiality concerning*

a *third-party* (Security for the OSI Reference Model: ISO 7498-2, ECMA, Bell/La Padula, etc.) [24] [28] [31] [3] [2] [9] [10] [12].

The *use of information bound to intended goals* is part of the economic dimension and an extension to traditional IT-security which has not been solved up to now. This extension requires the prevention of information misuse concerning the communication partners (not a third-party), i.e. restricting possible use of information.

5.2. The Framework

Our explanations of security requirements for information exchange in organisations show the need for considering organisations as socio-technical systems. We have defined three dimensions of information exchange in organisations. Humans act in the "real world" to reach personal goals and communicate with each other using an IT-system. This communication causes interpersonal relationships to be established. The representation of knowledge, facts or believes by the sender and the interpretation of the data by the recipient form the cross points between the IT-system and the "real world". Data processing and transmission are provided by the IT-system.

These dimensions build the structure of a comprehensive security framework (Fig. 3): Communication security, the security of content exchange and information security form the generic levels for secure information exchange.

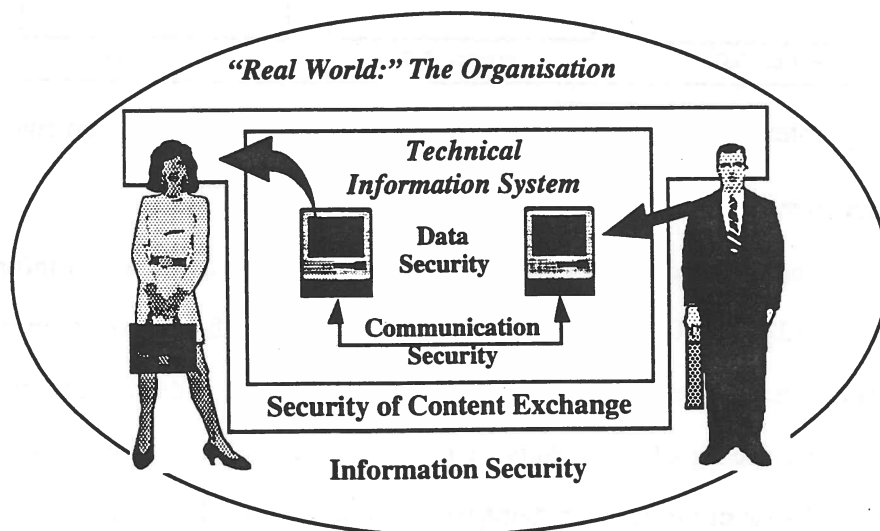


Fig. 3: Security levels of information exchange

The definition of security criteria for these levels provides the components of the security framework (Fig. 4). These criteria reflect generic security requirements according to the dimensions of information exchange in organisations (Fig. 2). In the following sections we briefly explain each of the criteria within the framework.

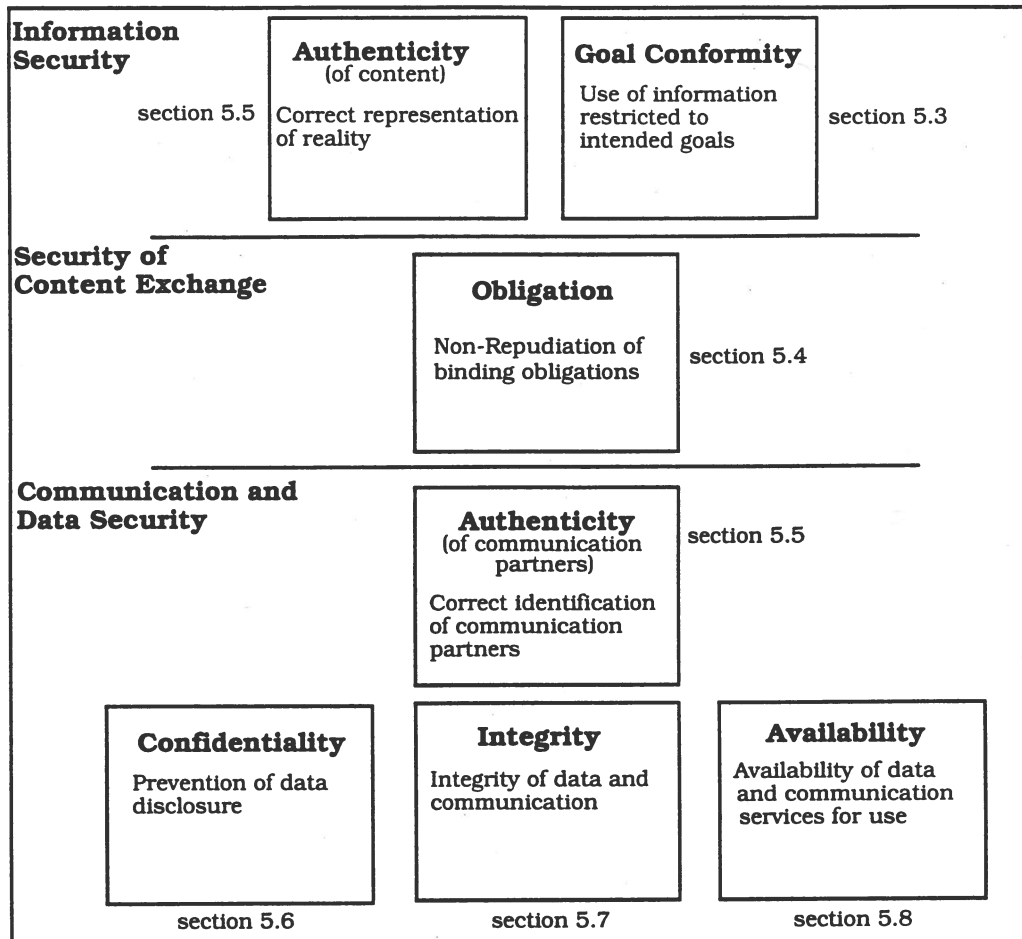


Fig 4: Security criteria of a comprehensive information exchange security framework

5.3. Goal Conformity

Goal conformity prevents the misuse of information, i.e. it ensures that the use of information is bound to the intended goals of the owner/sender. It considers the influence of information on the relationship between what humans think and their resulting goal-oriented actions (what they decide to do). To guarantee goal conformity, damage caused by the misuse of information must be prevented. Misuse of information damages the organisation when information enables an individual to act in a way that contradicts with the organisation's goals. Threats arising from the possible uses of information depend on the questions: What could anyone do with specific

information? What could a specific recipient do given their individual competencies? That means the recipient's interpretation of information is very important, especially when he or she has specific competence. Goal conformity extends the traditional IT-security criteria to meet the requirements of information exchange in organisations. It considers the economic relevance of information arising from human interactions.

5.4. Obligation

The exchange of information between communication partners includes social interaction. From the security point of view, we require that the obligations emerging from this interaction are binding to the communication partners, i.e. they accept the obligations. Their binding character is based on the kind of interaction, the legal systems, organisational policies, commitments or guidelines. For example, if a social interaction is documented as a legally binding contract, then obligations are provable and therefore enforceable in a social context [9, 10]. Like goal conformity, binding obligations are independent of the realisation of technical information systems and are an extension of traditional IT-security criteria.

5.5. Authenticity

Authenticity of communication partners is a widely accepted security criterion for information systems. It stems from the nature of interpersonal communication. Authenticity requirements encompass the authenticity of communication partners (as is the case of KERBEROS [27]) as well as the authenticity of a message's content, i.e. correspondence between representation and reality. Often the latter is not considered⁶ or discussed in the context of integrity. Authenticity of the content means that the representations can be validated according to the organisational "reality". In this sense, authenticity guarantees that valid decisions can be made based on the content of the system model.

5.6. Confidentiality

Confidentiality of data is a traditional IT-security criterion. It describes the state of being private or secret, i.e. only accessible by authorised people [21] [6] [2]. Unauthorised disclosure of data

⁶ Exceptions are the Clark/Wilson [4] and the Biba model [3].

is a violation of confidentiality. Confidentiality is ensured by data access controls or encryption in technical systems. Access controls are classified as either mandatory access controls that specify user privileges and object classifications (e.g. top-secret, private, public) or discretionary access controls that define discrete relationships between an active and a passive component [6, 2]. The more flexible techniques to access control are based on user-role descriptions [13, 32, 14] [7, 31]. To formally specify the underlying authorisations, various approaches are proposed in the literature [33] [28, 31]. Steinke, for example, uses a group security model which describes access rights through a teamwork-oriented organisational model [28]. Another way to ensure confidentiality is data encryption which uses cryptographic methods to preserve against the unauthorised interpretation of data. There are two major methods for data encryption: (a) symmetric encryption, i.e. encryption and decryption using the same key; and (b) asymmetric encryption, i.e. public and private keys for encryption and decryption respectively.

5.7. Integrity

Integrity, like confidentiality, is a traditional IT-security criterion [3]. The distinction between the integrity of the data and the correspondence between representation and reality is rarely considered when specifying security requirements. We call the correspondence between representation and reality the authenticity of content (see 5.5). Integrity, as we define it, relates to the internal state of data and depends on the technical realisation of the system in use, but not on the system's representation.

5.8. Availability

Availability is a state of existence or readiness for the resources and services (infrastructure, data, services) belonging to a technical information system. Availability is a basic requirement that is barely considered in existing security models [21].

6. Remarks to the Framework

The proposed framework shows the need for extensions of existing solutions for ensuring information security, e.g. measures to enforce or control the intended use of information ("goal conformity"). Therefore, it is necessary to understand the problems underlying "goal

conformity". This is a precondition for the development of technical measures to support this security criteria.

It is important to recognise the difference which we wanted to introduce in this paper namely, the difference between a classical (IT-oriented) interpretation of confidentiality and "goal conformity" as extension due to the characteristics of information exchange in organisations.

A comprehensive security concept needs all security criteria to be ascertained. Additionally the relationships and causalities must be understood as we have explained for data confidentiality and goal conformity. Homogeneity in selecting measures can only be achieved if we understand their co-operation and dependencies and therefore manage the related security criteria. The goal is to co-ordinate a set of possible measures (security resources) related to all the security criteria in a flexible way.

7. Managing Secure Information Exchange

Corresponding to the suggested framework, we briefly describe those requirements an information system in productive use should fulfil to achieve secure information exchange.

- *Classification of the economic potential of information to be exchanged* within a communication context, i.e. which activities of the communication partners are intended and which are possible? etc. Additionally, a classification of the communication partners is necessary: What organisational roles do they have? Which roles are possible based on their individual competence? etc.
- *Classification of the type of interpersonal interaction.* The type of interpersonal interaction refers to the obligations which arise out of this interaction.
- *Classification of the overall sensitivity of the data* for loss of confidentiality, integrity, and availability concerning a third party.
- *Anticipation of threats based on the previous classifications*, e.g. misuse of information, neglecting obligations, or disclosure of data. The probability of misuse can be anticipated, for instance, by using a model of goal-based human behaviour [20]. Although the risk of opportunistic behaviour is quite common in organisations, behaviour is not completely arbitrary, but governed by norms and rules. Acting depends on the situation, the individual's competence and on the available information to be exchanged. Usually the

goals of the communication partners overlap but are not congruent. Their differences must be anticipated to discover the conflicts of interests arising from the exchange of the information.

- *Selection of measures appropriate for preventing anticipated threats.* That is, the selection of security measures to be used for data access and transmission, e.g. document exchange; and the selection of security measures to prevent the misuse of information, to establish obligations and to guarantee the correctness of the content.

8. Conclusion

A comprehensive security concept for information exchange in organisations must take into consideration the interdisciplinary aspects of socio-technical systems, i.e. the technical information system within an organisation. This paper has focused information exchange as important part of fulfilling tasks in organisations, i.e. information influences/guides the human activities and behaviours in the organisation and the real world surrounding it. A primary focus of this paper has been to illustrate implications of this point of view to security.

We discussed the subjectivity of security (valuable objects, relevant threats and acceptable risks) and proceeded to explain the security relevant dimensions of information exchange. Then we described the components (security criteria) of a framework for information exchange. This security framework constitutes the basis for a systematic analysis of security requirements. According to its structure the framework guides the formal specification of requirements and the development of security concepts. As part of this process it shows the contribution of existing approaches to the overall security of information exchange [33] [28, 31]. Up to now existing approaches for secure information systems do not consider the economic dimension appropriately.

Further work will be done in this area by the OBS-project. The goal is to realise a security concept for information exchange in organisations. Therefore, measures for individual parts of the security framework must be developed, e.g. to enforce "goal conformity" and must be integrated with existing measures related to other criteria of the framework.

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